

CRAVECOIN

The Crypto Tracker

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Abstract: The Cravecoin application, a flutter project, has manually coded coin models for each cryptocurrency taking data from the CoinGecko API. The Cravecoin application sends an in-application notification to alert the user about the market whenever the cryptocurrency hits the same price set by the user. This price alert feature helps the users stay ahead of these changes and make as much as possible from the opportunities that come up with the upcoming future. The Cravecoin application is here to simplify your journey by providing an all-inclusive background and user-friendly platform to monitor, analyze, and optimize your cryptocurrency portfolio. With the Cravecoin application, you can effortlessly track the performance of your favorite cryptocurrencies in real time. Stay informed with personalized alerts and notifications tailored to your preferences. Set price alerts to receive instant updates on significant market movements, enabling you to make timely investment decisions. Whether the user is a seasoned trader or just starting, Cravecoin application is designed to serve to all the needs and simplify crypto journey. Start optimizing your cryptocurrency investments today with the Cravecoin application.

Keywords: Cravecoin, Cryptocurrency, Tracking, Real-time alerts, Flutter development, CoinGecko API, Price notifications, Investment optimization.

1. Introduction:

Cryptocurrency is a virtual or digital currency that uses cryptography for security. It operates independently without any interference of the central bank. Cryptocurrency can be transferred directly between any individuals around the world without the need for intermediaries like banks or financial institutions. Crypto investment is a fast-moving market that requires investors to act quickly. It fluctuates every second. One suitable investment can change your life. But the main issue is that one can't keep an eye on the market 24x7. A normal crypto investor loses many opportunities because he/she is not watching the market. In this era, tracking is made effortless with the Crypto Tracker App. Staying on top of your cryptocurrency investments is very crucial in this dynamically ever-evolving digital landscape. Cravecoin App is here to simplify your journey by providing an all-inclusive and user-friendly platform to monitor, analyze, and optimize your cryptocurrency portfolio. With the Cravecoin App, you can track the performance of your favorite cryptocurrencies in real time effortlessly. Gone are days of manually searching various platforms and exchanges for updated information. Our app consolidates data from trusted sources, giving you a holistic view of your investments right at your fingertips. Stay informed with personalized alerts and notifications tailored to your preferences. Set price alerts to receive instant updates on significant market movements, enabling you to make timely investment decisions. Our app keeps you up to date with the latest market prices, ensuring you stay ahead of the market in this fast-paced industry. Dive deep into historical data and trends to gain helpful insights of the market. User can easily analyze the performance of individual coins, identify patterns, and make decisions based on the data to maximize your returns. With the guarantee that your sensitive information is protected, you can track your crypto coins with peace of mind. Whether you are a seasoned trader or just starting, our app is designed to provide for all your needs and simplify your crypto journey. Start optimizing your cryptocurrency investments today with the Cravecoin App. Gain clarity, control, and confidence in managing your portfolio. Embrace the upcoming of the future in crypto tracking and get on a successful investment journey with the Cravecoin App.

2. Literature Review:

- [1] **"Crypto-Tracker Browser Extension"** by Gauri Rao, Sourabh Kumar Jha, Sadiq Hussain, Swastik Gupta, Pune. It's a browser extension which shows the values of the top ten cryptocurrencies in real time. Whenever the extension is clicked, the values get updated. The target browser for this application is 'Google Chrome'. The application is recognized by the browser via the "manifest.json" file. The application uses a background script which doesn't affect any content of the webpage whenever it is used.
- [2] **"Cryptocurrencies Performance Tracker and Data Visualization App"** by Saransh Bhardwaj, Sankalpa Basu, Mridul Pal, Gautam Buddha Nagar. The team has used programming languages such as HTML, CSS, and JavaScript to build the platform, and with that, they also used an API to get the cryptocurrency data. The platform they built gives the insight of the performance of cryptocurrencies with an attractive User Interface. The cryptocurrency data that is received is updated every day and includes price changes over a period of 24 hours and 1 week. This also includes the valuation of cryptocurrencies. The main objective of creating this platform was to provide easy access to crypto insights to the user. They have built their User Interface in such a way that it makes it easy for the user to explore every page without any problem.
- [3] **"Cryptocurrency Tracker Extension"** by Ms. Arti Bhatlawande, Mr. Akbar Shaikh, Prof. Sachin, Dept. of Computer Science & Engineering, SMSMPITR Akluj, Maharashtra. The main goal of creating this platform was to provide with easy access to cryptocurrency overviews for users. The team designed their user interface to make it easy for the user to explore each page seamlessly. The web application offers compactness, and it is user-friendly. The main moto of this project was to give information from the current to up to one year past crypto market and provide a simple user-friendly web application. It is a web application that displays the values of the top 10 cryptocurrencies in real time. The target browser for this application is 'Google Chrome'.
- [4] **"DELLA - Cryptocurrency Price Tracker"** by Shweta Saxena, Priyanshi Goyal, Prachi, Meerut. It makes use of encryption to provide the secure transfer and trade of digital docket across a distributed and decentralized network. These dockets can be

traded for cash at market prices. The primary goal of this study was to provide information to people or individuals who are unfamiliar with cryptocurrency. It's for those who wish to invest online while simultaneously dealing with economic operations like purchasing, selling, and trading. Della was shown as a cryptocurrency consultant in this study. It implies that it will happen.

[5] **“CryptoCurrency Tracker App”** by Shaikh Mohd Ashfaq, Vaibhav Prakash Palande, Akshaya Madan Samant, Ashish Kamlakar Naik, Assistant Professor of Computer Department, B.E Computer Engineering, Rizvi College of Engineering, Mumbai. The main goal of creating this application was to provide live cryptocurrency prices for users. The team designed their user interface to make it easy for the user to explore everything seamlessly.

[6] **“CRYPTONIK – Live Cryptocurrency Tracker”** by Deepesh Dandriyal in June 2023. Cryptonik was developed as a cryptocurrency tracker using React.js and the CoinGecko API. This paper describes the development of Cryptonik, its features, and its potential applications. Cryptonik was designed to be a simple and user-friendly cryptocurrency tracker. The user interface includes a dashboard that displays the value of various cryptocurrencies in real time. Users were able to select their preferred cryptocurrencies and were able to check detailed information about each one. Additionally, Cryptonik uses Firebase authentication to provide secure access to the users with their personalized dashboard.

[7] **“Cryptocurrency tracker for real-time market analysis and monitoring using React.js”** by Rahul Kumar, Aditya Mishra, Misbahul Mollick, Harish Prasad Chaurasiya, and Kunal Roy, Durgapur. The development process involves the utilization of APIs provided by cryptocurrency exchanges to fetch and update market data in real time. The integration of charting libraries enables users to visualize historical price data and identify patterns or trends that can aid in decision-making. Furthermore, the cryptocurrency tracker incorporates advanced features such as portfolio management, where users can track the performance of their cryptocurrency holdings and analyze their overall investment portfolio. Customizable alerts can also be set to notify users of specific market conditions or price movements, enhancing the tool's functionality and providing users with timely information. The research discusses the design and implementation of the cryptocurrency tracker, highlighting the key components, data flow, and user interface considerations. Additionally, it evaluates the performance and usability of the tracker through user testing and provides insights into potential areas for future enhancements and research.

[8] **“Real-time price tracker web application for cryptocurrency”** by Shruti C. Sharma, Dr. Pragati Richa Pandey, G. S. College of Commerce and Economics, Nagpur. They have used programming languages python to build the platform, and along with that, we have also used an API to get the cryptocurrency data. The platform they have built gave the insights about the performance of cryptocurrencies with an attractive GUI. The cryptocurrency data they receive is updated every day and includes price changes over 24 hours and 1 week. This also includes the valuation of cryptocurrencies. The main objective of creating this platform was to provide easy access to crypto insights to the user. They have built their UI in such a way that it makes it easy for the user to explore every page without any problem.

3. Research Methodology:

The research methodology for the Cravecoin project followed a structured approach. It emphasized a seamless blend of innovative technology and user-centered design to create an efficient cryptocurrency tracking application. The methodology focused on ensuring real-time data accuracy and an intuitive user interface.

Key stages of development were grounded in agile principles, allowing iterative improvements based on ongoing testing and user feedback. The technical choices reflected a commitment to delivering a robust, scalable, and user-friendly application. Below is a breakdown of the core technologies used in this project and their roles:

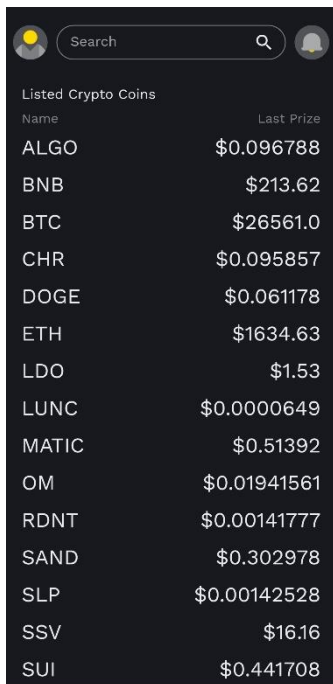
- **Flutter (Front-end Development):** Used to build the mobile interface, ensuring the app works smoothly on Android devices with a single codebase.
- **Firebase (Back-end & Database Management):** Integrated to store user data securely, manage user authentication, and synchronize information across devices in real-time.
- **CoinGecko API:** Employed to fetch real-time cryptocurrency data, including current prices, market capitalization, and 24-hour price changes.
- **Manually Created Models and Controllers:** Custom-built models and controllers to manage specific cryptocurrency data, ensuring smooth and fast integration with the APIs and allowing flexibility in handling data display and user interactions.
- **Dart Language:** Powers the logic and structure of the application, enabling efficient handling of asynchronous tasks such as fetching and updating data.
- **Flutter Local Notifications:** This package facilitates the in-app notification system, which alerts users when their selected cryptocurrencies hit the desired price points.

By leveraging these technologies, the Cravecoin app delivers a comprehensive solution to real-time cryptocurrency tracking, ensuring an engaging, efficient, and secure user experience.

4. Results:

The Cravecoin project achieved its objective of delivering a real-time cryptocurrency tracking application successfully. The app enables users to effortlessly monitor and manage their cryptocurrency investments, leveraging features such as real-time data updates, price alerts, and a user-friendly interface. Below are the key outcomes, supported by screenshots from the user interface design.

[1] **Home Page:** The home Page offers a streamlined, intuitive layout where users can view a list of tracked cryptocurrencies. Prices are updated in real time, allowing users to assess market conditions quickly. Cryptocurrencies are displayed in a tabular format with clear data points, such as current price, enabling users to interact with individual cryptocurrencies by clicking on their respective rows.

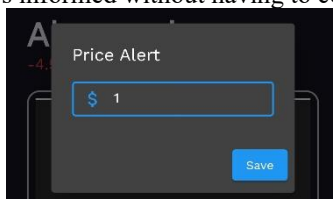


Name	Last Price
ALGO	\$0.096788
BNB	\$213.62
BTC	\$26561.0
CHR	\$0.095857
DOGE	\$0.061178
ETH	\$1634.63
LDO	\$1.53
LUNC	\$0.0000649
MATIC	\$0.51392
OM	\$0.01941561
RDNT	\$0.00141777
SAND	\$0.302978
SLP	\$0.00142528
SSV	\$16.16
SUI	\$0.441708

[2] **Crypto Page:** The Crypto Page provides detailed information on each selected cryptocurrency, such as market capitalization, total supply, and price changes over 24 hours. This page is very important for investors who want to dive deeper into the specific coin's performance metrics. Users can also easily set price alerts for specific currencies on this page.



[3] **Price Alert Setup:** A key feature of Cravecoin is the Price Alert Setup functionality. Users can specify a price for any designated given cryptocurrency. Once the price reaches that target, the app automatically sends a notification, ensuring the user stays informed without having to constantly monitor the market.

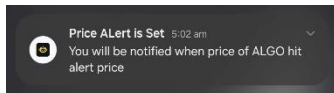


Price Alert

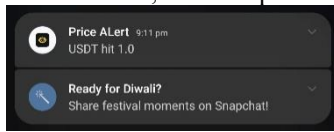
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[4] **On-Click Notification:** The On-Click Notification feature is designed to provide instant, in-app alerts when the specified price conditions are met. This notification system adds a layer of convenience for users by ensuring that they are instantly informed of important market changes without delay.



[5] **Price Alert Hit:** The Price Alert Hit functionality notifies users when their chosen cryptocurrency reaches the specified target price. This real-time alert is crucial for allowing users to take immediate action on their investments, capitalize on favorable market conditions, or avoid potential losses.



The outcome of this project highlights the app's capability to serve users with a responsive, data-driven experience. Through its seamless integration of APIs, user-friendly interface, and robust notification system, Cravecoin empowers users to make well-informed cryptocurrency investment decisions. The successful deployment of these features demonstrates the app's value as an indispensable tool for both novice and experienced investors alike.

5. Conclusion:

We are delighted to announce the successful development and deployment of our application, in line with the initial project proposal. Our team diligently adhered to the project's specifications and implemented a rigorous testing regimen to ensure a robust and reliable product for cryptocurrency investors. We firmly believe that our application represents a significant advancement in addressing the timing challenges prevalent in the cryptocurrency market. By providing investors with timely and relevant information, our solution aims to empower users to make informed decisions amidst rapidly changing market conditions. Moving forward, we are deeply committed to further enhancing our system by incorporating additional features and functionalities to improve the convenience and efficacy of crypto investments. Our dedication to excellence drives us to continually refine our solution to meet the evolving needs of the cryptocurrency community.

6. Reference:

- [1] **Crypto-Tracker Browser Extension** Gauri Rao, Sourabh Kumar Jha, Sadiq Hussain, Swastik Gupta, Pune
- [2] **Cryptocurrencies Performance Tracker and Data Visualization App** Saransh Bhardwaj, Sankalpa Basu, Mridul Pal, Gautam Buddha Nagar
- [3] **Cryptocurrency Tracker Extension** Ms. Arti Bhatlawande, Mr. Akbar Shaikh, Prof. Sachin, Maharashtra
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- [8] **Real-time price tracker web application for cryptocurrency** Shruti C. Sharma, Dr. Pragati Richa Pandey, Nagpur.
- [9] https://youtu.be/j-LOab_PzzU?si=x3q7JvpYCsOSDvph - Master Flutter in Just 8 Hours | Full Course in Hindi by HindiCodepur for learning Flutter language and its concepts.